Advanced information, computation, communication I
EPFL - Fall semester 2026-2027

Week 5

October 9, 2026

Exercise 1. Let A and B be sets. For each of the following cases, determine whether the given object defines a relation from A to B. Which of the following statements are correct?

◯ Let $A = \{1, 2, 3\}$, $B = \{a, b\}$, and

$$R = \{(1, a), (2, b), (3, a)\}.$$

Then R is a relation from A to B.

◯ Let $A = \{1, 2, 3\}$, $B = \{a, b\}$, and

$$R = \{(1, a), (2, b), (4, a)\}.$$

Then R is a relation from A to B.

◯ Let $A = \mathbb{Z}$, $B = \mathbb{Z}$, and

$$R = \{(x, y) \in \mathbb{Z}^2 : y = 0\}.$$

Then R is a relation from A to B.

◯ Let $A = \mathbb{R}$, $B = \mathbb{R}$, and

$$R = \{x \in \mathbb{R} : x^2 < 1\}.$$

Then R is a relation from A to B.

Exercise 2. For each relation below, determine which of the following properties it satisfies:

- reflexive;
- symmetric;
- anti-symmetric;
- transitive.

(a) On $\mathbb{Z}$, define $xRy \iff x - y$ is even.
(b) On $\mathbb{Z}$, define $xRy \iff x < y$.
(c) On $\mathbb{Z}$, define $xRy \iff x \leq y$.
(d) On $\mathbb{R}$, define $xRy \iff |x - y| \leq 1$.
(e) On $\mathbb{Z}$, define $xRy \iff xy \geq 0$.
(f) On $\mathbb{N}$, define $xRy \iff x \mid y$

Exercise 3. Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (3, 4)\}$.

1. Which of the equivalence-relation properties fail for R?
2. We want to add the minimum possible number n of ordered pairs to R so that it becomes an equivalence relation. Determine n and pair(s) that should be added.

Exercise 4. A binary relation R on a set A is said to be quasi-reflexive if every element that is part of the relation is related to itself: $\forall a \in A$ and $\forall b \in A$, we have $(a, b) \in R \rightarrow [(a, a) \in R \wedge (b, b) \in R)]$. Let us assume that we have a non-empty binary relation R on a set A that is both symmetric and anti-symmetric. Then R is:

- ◯ Always reflexive and always quasi-reflexive
- ◯ Always reflexive but not always quasi-reflexive
- ◯ Not always reflexive but always quasi-reflexive
- ◯ Not always reflexive and not always quasi-reflexive

Exercise 5. Draw the Hasse diagram for divisibility on the set:

1. {1, 2, 3, 4, 5, 6, 7, 8}
2. {1, 2, 3, 5, 7, 11, 13}
3. {1, 2, 4, 8, 16, 32, 64}

Exercise 6. Given two posets, $(S, \preceq_1)$ and $(T, \preceq_2)$, show that $(S \times T, \preceq)$ forms a poset when ordered by the relation $\preceq$ defined as: $(s, t) \preceq (u, v)$ if and only if $s \preceq_1 u$ and $t \preceq_2 v$.

Exercise 7. Find the lexicographic ordering of these n-tuples:

1. (1, 1, 2), (1, 2, 1)
2. (0, 1, 2, 3), (0, 1, 3, 2)
3. (1, 0, 1, 0, 1), (0, 1, 1, 1, 0)

Exercise 8. For each of these sequences find a recurrence relation satisfied by this sequence. (The answers are not unique because there are infinitely many different recurrence relations satisfied by any sequence.)

1. $a_n = 3$
2. $a_n = 2n$
3. $a_n = 2n + 3$
4. $a_n = 5^n$
5. $a_n = n^2$
6. $a_n = n^2 + n$
7. $a_n = n + (-1)^n$
8. $a_n = n!$

Exercise 9. Find the solution to each of these recurrence relations and initial conditions.

1. $a_n = 3a_{n-1}$ and $a_0 = 2$
2. $a_n = a_{n-1} + 2$ and $a_0 = 3$
3. $a_n = a_{n-1} + n$ and $a_0 = 1$

4. $a_n = a_{n-1} + 2n + 3$ and $a_0 = 4$
5. $a_n = 2a_{n-1} - 1$ and $a_0 = 1$
6. $a_n = 3a_{n-1} + 1$ and $a_0 = 1$
7. $a_n = na_{n-1}$ and $a_0 = 5$
8. $a_n = 2na_{n-1}$ and $a_0 = 1$

Exercise 10. Suppose that the number of bacteria in a colony triples every hour.

1. Set up a recurrence relation for the number of bacteria after n hours have elapsed
2. If 100 bacteria are used to begin a new colony, how many bacteria will be in the colony in 10 hours?

Exercise 11. Solve the following recurrence relation using generating functions: $a_n = 4a_{n-1} - 4a_{n-2}$, $n \geq 2$, with $a_0 = 1$ and $a_1 = 4$.

Exercise 12. We have seen in the course that we can add, multiply, and, if the divisor has a non-zero constant term, divide formal power sums (FPS). In addition we have seen that we can take the formal derivative of a FPS. The key for all those operations is that we can compute the n-th coefficient of the result using only a finite number of basic operations. There are many other operations one can define.

Let $A(x) = \sum_{n\geq 0} a_n x^n$ and $B(x) = \sum_{n\geq 0} b_n x^n$ be two FPS. Under what condition is the composition $A(B(x))$ well defined? Compute the first 3 coefficients.